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DIVERGENT CRETACEOUS PALEOMAGNETIC POLE POSITION FOR THE SOUTHERN CALIFORNIA BATHOLITH, U.S.A.

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Paleomagnetic measurements on 18 stable sites in plutonic rocks from the Cretaceous Southern California Batholith yield an average direction of magnetization of 3.0°, 49.5° (declination, inclination) and a mean paleomagnetic pole of 85.5°N, 22.5°E (latitude, longitude). All sites are of Normal polarity. The mean direction of magnetization for the Batholith diverges from the expected Cretaceous direction by some 26° in declination and 11.5° in inclination. This divergence can be accounted for by reference to the mechanics of transform faulting between the North American and Pacific plates.

1. Introduction

This investigation originated as a paleomagnetic study of the San Marcos Gabbro phase of the Southern California Batholith [1]. In light of early results it was decided to extend the investigation to other phases of the batholith. This later work confirmed our original results and leads us to suggest the possibility of a large post-Cretaceous tectonic rotation for a part of the Southern California area southwest of the San Andreas fault. Recent evidence suggests that at least part of the rotation may be as late as post-Pliocene [2].

The Southern California Batholith is a composite granite pluton exposed from near San Bernardino, California, southward for approximately 550 km (fig. 1) [3, 4]. Phases of the batholith have been dated at 121 my by Herzog and Pinson [5], using Rb–Sr, and at 100 my by Baadsgaard et al. [6], using K–Ar. These radiometric dates support Larsen's [3] assignment of the Southern California Batholith to the mid-Cretaceous on the basis of geological field relations.

2. Methods of investigation

A total of 202 samples was collected from 30 sites scattered throughout the northern part of the batholith (fig. 1). Seventeen sites were located in 10 separate bodies of San Marcos Gabbro, while the remaining 13 sites were in more felsic rock, ranging in composition from diorite to quartz monzonite. The sampling encompassed an area of roughly 5000 km^2. Samples in the form of cores 2.35 cm in diameter and between 2.5 and 12.5 cm long were drilled and oriented in the field. Usually 5 to 8 cores were drilled at each sampling site. Deep weathering restricted sampling to artificial exposures, mainly highway cuts and quarries.

Directions and intensities of the remanent magnetization were measured at two different laboratories on

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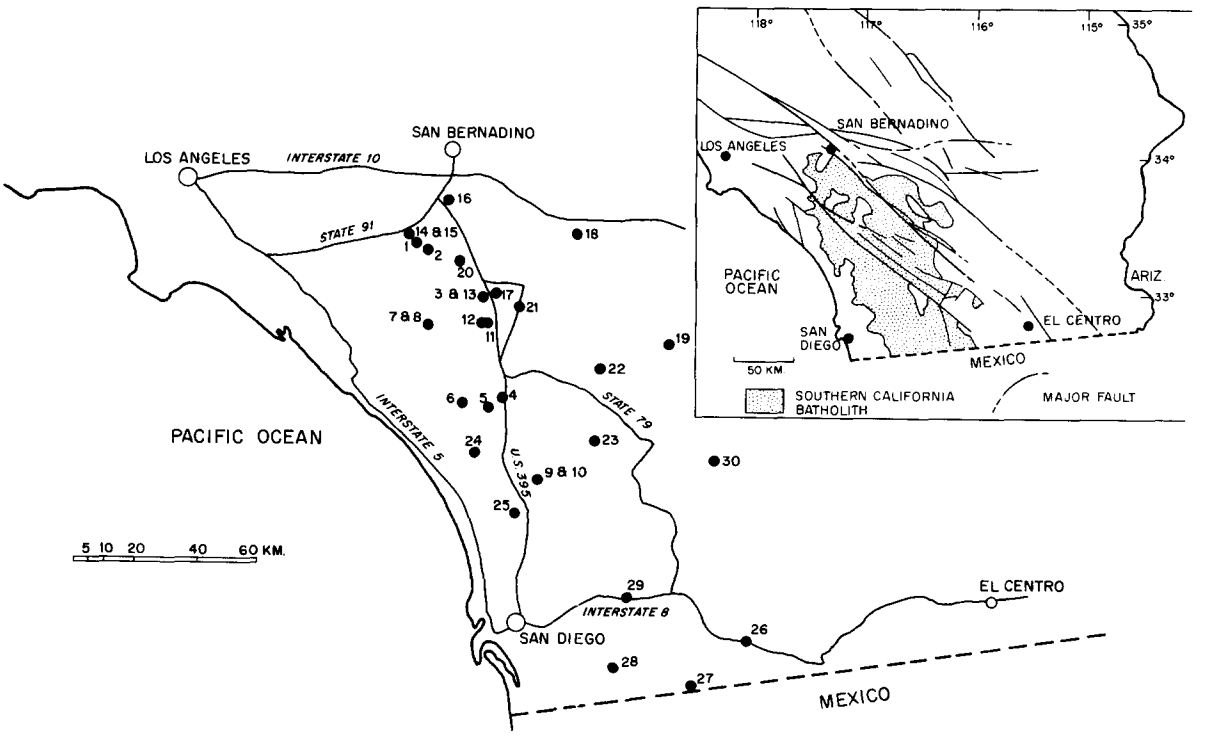


Fig. 1. Sampling localities. Insert shows extent of Southern California Batholith exposures in California.

Table 1
Paleomagnetic data for the Southern California Batholith

Site	Rock type	N^a	Dec.	Inc.	$\tilde{H}^b$	K^c	$\alpha_{95}{}^d$	Lat.[e]	Long.[e]
4	San Marcos Gabbro	4	4.5°	45.5°	100 Oe	55.4	12.4°	82.5°N	30.0°E
6	San Marcos Gabbro	4	7.0	52.0	50	81.0	10.3	84.0	340.0
7	San Marcos Gabbro	4	21.0	57.0	50	33.9	16.0	72.5	313.0
8	San Marcos Gabbro	5	348.5	66.5	50	14.4	20.8	72.5	217.0
10	San Marcos Gabbro	5	1.5	44.0	50	532.2	3.3	82.5	52.5
11	San Marcos Gabbro	6	4.5	70.0	50	35.1	11.5	69.5	250.5
13	San Marcos Gabbro	5	1.5	47.0	50	62.1	4.8	84.0	50.0
14	San Marcos Gabbro	5	3.0	35.5	100	182.6	5.7	75.5	51.5
15	San Marcos Gabbro	5	4.5	35.5	100	585.0	3.2	75.5	45.5
25	San Marcos Gabbro	10	11.5	42.0	100	135.4	4.2	76.5	12.5
28	San Marcos Gabbro	8	1.5	45.0	100	16.6	14.0	83.0	53.0
Mean	San Marcos Gabbro	11	4.5	49.0		44.4	6.9	85.5°N	0.5°E
16	Granodiorite	6	339.0	43.5	100 Oe	71.3	8.0	70.0°N	134.0°E
20	Diorite	7	4.0	44.0	100	14.9	16.1	81.5	38.5
21	Granodiorite	8	6.5	49.5	100	353.0	3.0	83.5	0.0
23	Tonalite	7	355.0	64.0	100	17.0	15.1	77.0	227.5
24	Tonalite	8	12.5	47.0	100	51.7	7.8	78.0	355.5
27	Tonalite	7	1.5	57.5	100	18.4	14.4	85.0	255.5
29	Tonalite	7	6.0	38.0	100	12.8	17.5	77.0	38.0
Mean	Non-gabbro sites	7	1.0	49.5		49.7	8.7	87.5°N	46.0°E
Mean	All sites	18	3.0	49.5		48.5	5.0	85.5°N	22.5°E

(a) Number of individuals in calculation. (b) Peak value of demagnetizing field. (c) Precision parameter [7]. (d) Radius of circle of 95% confidence [7]. (e) Coordinates of virtual geomagnetic pole.

replicas of the U.S. Geological Survey spinner magnetometer [13]. Systematic magnetic cleaning by the alternating field technique was applied to all samples. For sites 1 through 15 (table 1), the optimum field for magnetic cleaning was chosen by subjecting two samples from each site to stepwise demagnetization up to a peak value of 400 Oe. The remaining samples from each site were then magnetically cleaned in whatever field produced the smallest angle between the RM direction of the test samples from that site. For sites 16 through 30, all samples were subjected to stepwise magnetic cleanings, and the cleaning level yielding minimum dispersion was selected as the best estimate of the true site-mean direction of magnetization. As can be seen from table 1, optimum cleaning fields were unusually low, ranging from 50 to 100 Oe. Test specimens have been demagnetized in higher fields (up to 2000 Oe) but no further reduction of dispersion nor systematic change in direction has been detected.

Table 1 and fig. 2 give the results for all sites with circles of 95% confidence [7] of less than 25°. Sites with larger confidence circles in general show evidence of marked magnetic instability, several being statistically random, by the test of Watson [8]. These sites will not be considered further. In addition, site 12 has been excluded because there is some question as to whether or not the block in which it is located is in place. Finally, seven individual samples were excluded arbitrarily from sites otherwise judged acceptable, on the grounds that their directions diverged by large angles from otherwise tight groupings of directions. After these exclusions there remained 111 samples from 18 sites from which to estimate the mean direction of remanent magnetization of the northern part of the Southern California Batholith.

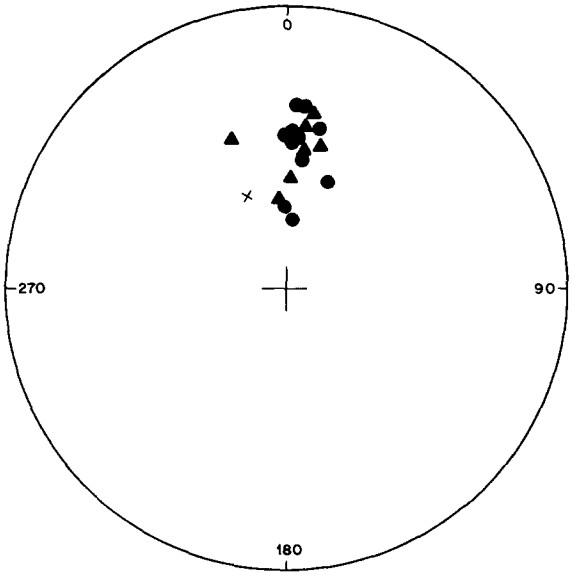


Fig. 2. Equal area projection of mean directions of magnetization for stable sites. Circles are gabbro sites, triangles are non-gabbro sites. All points in the lower hemisphere. × shows direction of the Cretaceous geomagnetic field expected in the sampling area on the basis of other paleomagnetic studies from the north American craton.

3. Discussion

Figs. 1 and 2 show no obvious correlation between magnetic direction and location among the various sampling sites. This implies that there have been no large tectonic rotations of any single site, or group of sites, relative to the rest of the batholith. Furthermore, the tight grouping of site-mean directions shown in fig. 2 indicates that little if any movement of the pole relative to the sampling area can have oc-

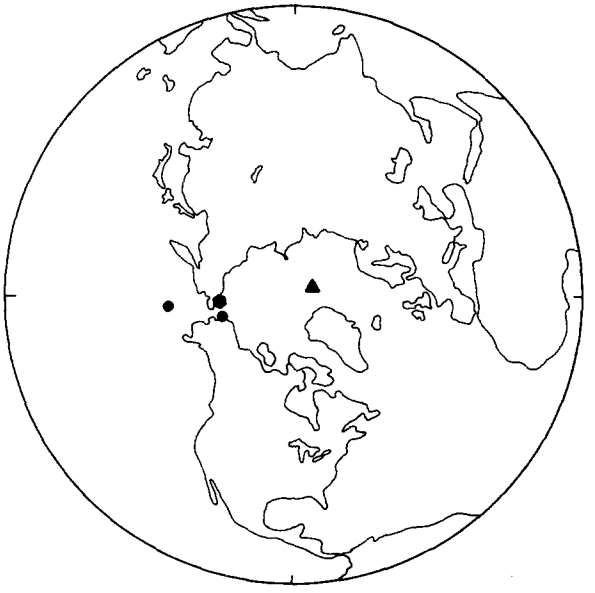


Fig. 3. Paleomagnetic pole positions. Circles, poles for Sierra Nevada batholith [12]; triangle, Southern California Batholith pole; hexagon, mean Cretaceous pole for cratonic North America [9].

curred while magnetization was being acquired, while on the other hand the large size of the sampling area and the probable slow cooling of these plutonic rocks should assure that the magnetization process extended over enough time to permit the averaging out of non-dipole elements of the geomagnetic field. Thus the mean pole listed in table 1 should represent a valid Cretaceous paleomagnetic pole, suitable for direct comparison with other North American poles of the same age.

Fig. 3 is a plot of pole positions for the Southern California and Sierra Nevada batholiths. Also shown is the mean Cretaceous paleomagnetic pole for cratonic North America calculated by Beck and Noson [9]. Clearly the Sierra Nevada poles are part of the cratonic group, whereas the Southern California Batholith pole is not. The mean Cretaceous cratonic pole (69°N, 175°W) would give a geomagnetic field direction with a declination of about 337° and an inclination of about 61° in the area sampled in this investigation. The corresponding direction actually measured is 3.0° and 49.5°. Thus there is a difference of about 26° in declination and 11.5° in inclination between the paleogeomagnetic field direction predicted from the known location of the pole relative to North America during the Cretaceous and the direction measured in batholithic rocks of Cretaceous age in Southern California. This is almost precisely the discrepancy noted for muds of Pliocene age from nearby Baja California [2].

Atwater [10] in her discussion of the tectonics of western North America in terms of the plate-tectonics model [11] introduces the concept of the western United States as a "broad transform fault zone." By this she apparently means that much of the Cordillera has behaved as a zone of simple shear across which relative motion between the Pacific and North American plates has been distributed. One likely consequence of such a tectonic pattern is that blocks of rock caught up in this zone of transform faulting might be translated northward and rotated clockwise, relative to the North American craton. Certainly the Sierra Nevada block has experienced little such relative motion since its granitic rocks became magnetized in the late Mesozoic, but available paleomagnetic evidence from Southern California appears to be compatible with such a model. Further discussion of this somewhat speculative concept should await additional paleomagnetic data, in particular from Baja California.

Acknowledgements

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